

# Alvin Tan

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PhD Candidate at UC Berkeley with broad research experience spanning reinforcement learning, embedded sensing, and energy infrastructure. Skilled in building end-to-end systems from simulation to deployment, mentoring junior researchers, and communicating through publications, presentations, and code releases.

## Education

**UC Berkeley** -- Ph.D. Candidate in EECS -- Advisor Prof. Prabal Dutta Expected Dec 2026

*Honors: NSF Graduate Research Fellowship Program (HM), EECS Outstanding TA Award (Top 1%)*

**Northwestern University** -- B.S. in Comp Eng, Econ, and Math GPA 3.99/4.00 | June 2020

*Honors: summa cum laude, Goldwater Scholarship (HM), ECE Outstanding Graduating Senior (#1)*

## Technical Skills

**Machine Learning / Deep Learning** (PyTorch, TensorFlow, CUDA); **Programming** (Python, C, C++); **Robotics** (ROS2, kinematics, coordinate transforms, computer vision, planning); **Controls** (PID, LQR); **Simulation** (Python, MATLAB); **Embedded Systems** (ARM Cortex M4 / M0+; I2C, SPI, GPIO; RTOS); **Debugging/Profiling** (oscilloscope, logic analyzer); **Networking** (BLE, Wi-Fi, Ethernet; REST, mDNS); **Knowledge Dissemination** (8 publications, 5 presentations, 5 code, 2 hardware demos, 1 short film)

## Main Research Projects

**Reinforcement Learning for Uncertainty-Aware Social Robot Navigation** Aug 2021 - June 2024

- Advanced the state-of-the-art in social robot navigation by publishing and presenting at ICRA'24, building on 21 papers about robot navigation, proxemics, pathfinding, and uncertainty quantification
- Reduced human-robot collisions by 80% in hard scenarios, by training a reinforcement-learning model (100k trainable params) with pedestrian uncertainty estimation and custom reward function
- Demonstrated successful sim-to-real transfer of learned behavior, where robot-pedestrian distance increases with uncertainty, navigating a 2-wheeled robot with computer vision and motor controls

**Automatic Transmission-Reduction for Live, Low-Data-Rate Data Sensors** June 2023 - present

- Advanced the SotA on live data streaming for user applications by publishing in HumanSys'25, building on 66 papers about real-time timeseries compression, data loss, and data visualization
- Achieved 2x higher compression over state-of-practice, by decomposing 21 compression algorithms into 12 functional categories and optimizing across 15 environmental, health, and industrial datasets
- Mentored 3 undergraduates on designing modular libraries and APIs, with 2 papers in progress

## Internships

**Resource-Optimization for Battery- and PV-enabled DC Microgrids** June 2025-present

*Lawrence Berkeley National Laboratory (Dr. Daniel Gerber)*

- Advanced the state-of-the-art in panel capacity management by publishing in ICDCM'26 and MDPI *Energies*, building on 70 papers about DC microgrids, smart house panels, and power management
- Assessed that a 10 kWh battery could reduce required residential panel capacity by 80% (70A to 15A) and increase local solar utilization by 3000 kWh/year, using a custom-built DC microgrid simulator

**Foundation Models for Atmospheric Science**

May - Aug 2023

*Johns Hopkins University Applied Physics Laboratory (Dr. Jennifer Sleeman)*

- Identified 3 failure modes of a climate-forecasting foundation model when generalizing across spatial resolutions and meteorological datasets, by designing and conducting novel fine-tuning experiments
- Transferred knowledge presenting to 50 researchers and advising a colleague on similar evaluations

## Selected Publications

\*co-first authorship

**Alvin Tan**, D. Gerber, P. Dutta. Less is the New More: On Reducing Panel Capacity Requirements using PV- and Battery-Supported DC Microgrids. *MDPI Energies*, 2026. DOI: 10.3390/en19174051.

**Alvin Tan**, P. Dutta. Good, but Not That Good: An Honestly-Noisy Visualization of Low-Fidelity Data Streams. *Proc. HumanSys'25*. DOI: 10.1145/3722570.3726882.

**Alvin Tan**. A Sum of Parts: Contemplating the Value of Life and Gentleness through Technology. Short film presented in the Arts in Robotics gallery exhibition at ICRA'25.

S. Pohland\*, **Alvin Tan**\*, P. Dutta, C. Tomlin. Stranger Danger! Identifying and Avoiding Unpredictable Pedestrians in RL-based Social Robot Navigation. *Proc. ICRA'24*. DOI: 10.1109/ICRA57147.2024.10610413.